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Prenatal Exposure to Per- and Polyfluoroalkyl Substances (PFASs) and Association between the Placental Transfer Efficiencies and Dissociation Constant of Serum Proteins−PFAS Complexes

Ke Gao,[†,‡,⊥] Taifeng Zhuang,[§,⊥] Xian Liu,[†] Jianjie Fu,[*,†] Jingxing Zhang,[†,∥] Jie Fu,[†,‡] Liguo Wang,[†,‡] Aiqian Zhang,[†,‡,∥] Yong Liang,[∥] Maoyong Song,[†] and Guibin Jiang[†,‡]

[†]State Key Laboratory of Environmental Chemistry and Ecotoxicology, Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences, Beijing, China

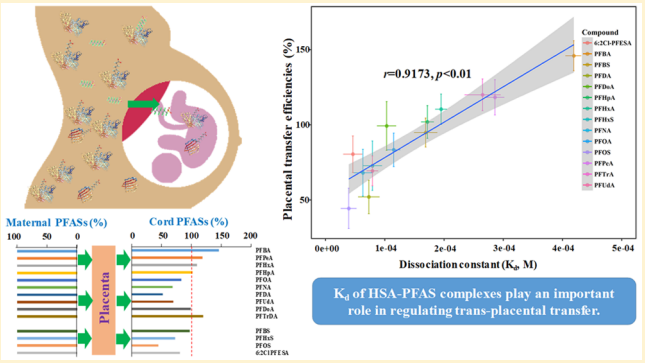
[‡]College of Resources and Environment, University of Chinese Academy of Sciences, Beijing, China

[§]Department of Pediatrics, Beijing Obstetrics and Gynecology Hospital, Capital Medical University, Beijing, China

[∥]Institute of Environment and Health, Jianghan University, Wuhan, Hubei, China

Supporting Information

ABSTRACT: Information on placental transfer and adverse outcomes of short-chain per- and polyfluoroalkyl substance (PFASs) is limited, and factors responsible for PFAS placental transfer are still unclear. In the present study, concentrations of 21 PFASs were analyzed in 132 paired maternal and cord serum samples collected from residents in Beijing, China, and the placental transfer efficiency (PTE) of each PFAS was calculated. PTEs of short-chain perfluoroalkyl acids (PFAAs), including PFBA (146%), PFBS (97%), PFPeA (118%), and PFHxA (110%), were first reported, and a complete U-shaped trend of PTEs from C4 to C13 of perfluoroalkyl carboxylic acids (PFCAs) was obtained. Positive association between maternal weight and PTE of perfluorooctanesulfonate (PFOS) ($p < 0.05$) and negative association between maternal PFBA concentration and birth length ($p < 0.01$) were observed. Using in vitro experiments, we further determined equilibrium dissociation constants (K_ds) of human serum albumin (HSA)−PFAS complexes ($K_{d\text{-}HP}$), serum proteins−PFAS complexes ($K_{d\text{-}SP}$), and liver-fatty acid binding protein (L-FABP)−PFAS complexes ($K_{d\text{-}LP}$) and found that they were all significantly correlated with PTEs of PFASs. The correlation coefficient was 0.92, 0.89, and 0.86, respectively ($p < 0.01$ in all three tests), suggesting that K_ds of protein (serum)−PFAS complexes can play an important role in trans-placental transfer of PFASs in human and $K_{d\text{-}HP}$ plays a pivotal role.



INTRODUCTION

Per- and polyfluoroalkyl substances (PFASs) are a broad class of synthetic substances consisting of a fluorinated carbon chain and one or more functional groups, and they are extensively used in industrial,[1] commercial,[2] and consumer products.[3] Perfluorooctanesulfonic acid (PFOS) and perfluorooctanoic acid (PFOA) are two representative perfluoroalkyl acids (PFAAs) that are ubiquitously detected in the environment and in humans due to their large historical usage, high stability, and bioaccumulative behavior.[4−6] Since PFOS and PFOA have been under environmental scrutiny and global regulation,[7,8] their alternatives, compounds with similar fluorinated chain structures, such as short-chain PFAAs[9] and polyfluorinated ethers,[10,11] are being produced. These alternative PFASs have been found in human tissues, some even with increasing trends, suggesting ongoing exposure to emerging PFASs.[12]

PFASs are proteinophilic compounds, and they are mostly distributed in serum.[13,14] Previous studies have shown that PFASs could transfer from the mother to the fetus during pregnancy.[15−17] The potential adverse health effects associated with PFAS, such as neurobehavioral problems,[18,19] immune suppression,[20,21] and increased adiposity,[22,23] are currently focused on children. Studies also suggested that PFASs could result in lower birth weight,[24−26] and prenatal PFAS exposure was associated with changes in circulating lipid profile and alanine aminotransferase (ALT) levels in children.[27] Placental transfer efficiency (PTE) is the ratio of PFAS concentration in cord blood to that in maternal blood, which is an important marker in assessing fetal PFAS burden, and PTE is considered to be affected by the functional group and the fluorinated chain length of PFASs.[15,16] It has been observed that the PTEs of

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PFASs showed a U-shaped trend with increasing perfluoroalkyl chain lengths,[16,17,28] and this has been attributed to the different binding affinities of PFASs to proteins. However, to our knolwedge, no observational study has examined the correlation between placental transfer and experimental measurements of dissociation constant (K_d). Moreover, some researchers have expressed concern on the potential toxicity of short-chain PFAAs,[9,29] but studies investigating short-chain PFAAs in paired maternal and cord serum are scarce due to limitations in analytical methods (e.g., high method limits of detection, poor peak shape, and blank contamination). To our best knowledge, there were no studies on PTEs of short-chain perfluoroalkyl carboxylic acids (PFCAs) (C4−C6) and perfluorobutanesulfonic acid (PFBS). Our lab developed sensitive online TurboFlow SPE HPLC-MS/MS analytical methods for PFASs without chain-length discrimination recently,[30,31] providing the feasibility of the calculation of PTEs of short-chain PFAAs. The present study aimed to fill the data gaps in PTEs for short-chain PFAAs.

Generally, there are two main modes of entry in the human body for PFASs, namely, through ingestion and inhalation.[32,33] PFASs could stay in the human body as a consequence of enterohepatic recirculation.[34] PFASs have strong binding affinity to serum proteins,[13,14] resulting in their prominent presence in human serum compared to specific tissues.[35] Most PFASs in serum are in the bounded form, while only a small fraction exist as free molecules. Human serum albumin (HSA) is the major protein in serum, accounting for 60% of total plasma protein content;[36] it is also the major binding protein for PFASs.[37] Many previous studies have employed HSA to demonstrate the accumulation and toxic mechanism of PFASs in humans.[38−40] Although liver-fatty acid binding protein (L-FABP) is mainly present in the liver, it is also one of the major proteins in serum with a concentration of 9.46 μg/L (4.5−17.5 μg/L).[41] Due to similar structure patterns between PFASs and fatty acids, L-FABP is also an important binding protein for PFASs. Moreover, the binding between L-FABP and PFASs is suggested to potentially influence the uptake and transport of fatty acids and might further interfere with glycolipid metabolism.[14,42] PFAS−protein complexes, as macromolecules, are difficult to pass through the placental barrier; thus, only free PFASs can cross the placental barrier.[17,43] Hence, the free state concentration of PFASs affects the transport efficiency, and binding ability might be a critical factor to regulate PTE. Binding affinity is generally measured and quantified by an equilibrium K_d, which is the inverse of the binding affinity (K_a). Thus, we hypothesized that K_d is one of the decisive factors for PTE of PFASs in humans. The literature has presented a hypothesis that the binding affinity to albumin and/or L-FABP is associated with U-shaped trend for PTEs of PFASs.[16,17] However, no experimental evidence was reported in an observational study. Therefore, we adopted K_d as the metric for the affinity of PFASs binding to protein and hypothesized that K_d is one of the decisive factors for PTE of PFASs in humans.

We collected 132 paired maternal and cord serum samples in the present study, and the main objectives were to (1) investigate the occurrence of 21 PFASs and determine their PTEs, (2) explore the possible associations among maternal serum PFASs, cord serum PFASs, and infant birth outcomes, as well as the possible associations between individual characteristics and PTEs of PFASs, and (3) determine the K_d of HSA−PFAS complexes ($K_{d\text{-}HP}$), K_d of L-FABP−PFAS complexes ($K_{d\text{-}LP}$), and K_d of serum proteins−PFAS complexes ($K_{d\text{-}SP}$) to find out the key factor in maternal-fetus PFAS transfer.

MATERIALS AND METHODS

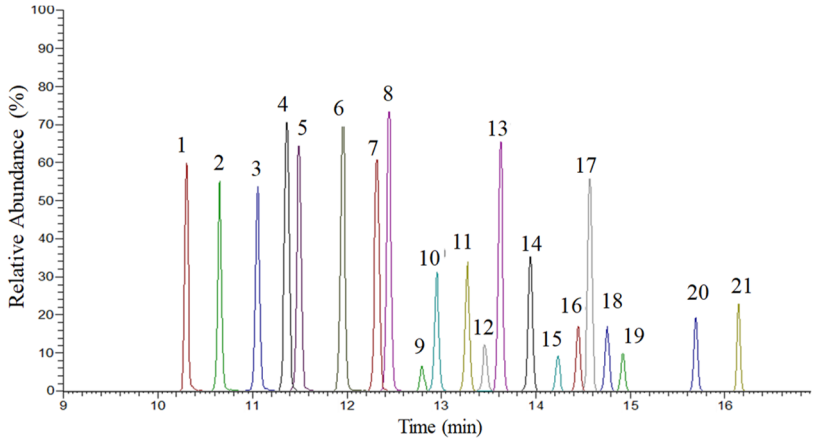
Study Design and Sample Collection. The study was approved by the ethics committee of the Affiliated Hospital of Capital Medical University and carried out in accordance with the approved guidelines. The participants were fully informed of the nature and purpose of the study, and consent forms were obtained. We issued questionnaires to women who visited regional hospitals for delivery from 2015 to 2016 at the Affiliated Hospital of Capital Medical University at a rate of 2−3 persons per day. To investigate prenatal exposure and exclude other confounding factors, volunteers were in healthy condition and required to be lived in Beijing for at least 1 year and gave birth to a single live-born child without congenital anomalies. Finally, 132 pregnant women were recruited. Maternal blood samples were collected 1 or 2 days before delivery, while cord blood samples were collected immediately after delivery. The blood samples were centrifuged at 3000 rpm for 15 min immediately after collection in the hospital. Then serum was transferred to 3 mL polypropylene containers and stored at −80 °C prior to analysis. Demographic data on the mothers and infants were obtained from the questionnaire or the medical records at delivery (Supporting Information (SI) Table S1). Demographical characteristics information about three pairs of paired mother−infant and two mothers were missing. The mean ± standard deviation (SD) age of the participants at delivery was 31.0 ± 3.74 years, the mean pregnancy body mass index (BMI) was 27.1 ± 3.38 kg/m^2, and the mean gestational age was 38.8 ± 1.32 weeks. In all, 77% of pregnant women were primiparous, and 89% of fetuses had a normal gestational age (37−40 weeks). All participants did not smoke or consume alcohol during pregnancy and did not have an occupational exposure experience to PFASs. As for infants, the number of female infants (n = 63) was almost equal to that of males (n = 66). The birth weight, birth length, and ponderal index (PI) of the infants were 3478 ± 431 g, 50.0 ± 1.0 cm, and 27.7 ± 2.76 kg/m^3, respectively.

Laboratory Analysis of PFASs. In total, 21 targeted PFASs (including perfluorobutanoic acid (PFBA), perfluoropentanoic acid (PFPeA), perfluorohexanoic acid (PFHxA), perfluoroheptanoic acid (PFHpA), PFOA, perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA), perfluoroundecanoic acid (PFUdA), perfluorododecanoic acid (PFDoA), perfluorotridecanoic acid (PFTrA), perfluorotetradecanoic acid (PFTeA), perfluorohexadecanoic acid (PFHxDA), perfluorooctadecanoic acid (PFODA), perfluorobutanesulfonic acid (PFBS), perfluorohexanesulfonic acid (PFHxS), perfluoroheptanesulfonic acid (PFHpS), PFOS, perfluorodecanesulfonic acid (PFDS), perfluorooctane sulfonamide (PFOSA), 6:2 Cl-PFESA, 8:2 Cl-PFESA) were analyzed in the present study. Our previously developed online SPE-HPLC-MS/MS method,[31] consisting of a UltiMate 3000 DGLC high-performance liquid chromatography (ThermoFisher Scientific, USA) and a TSQ Quantiva triple quadrupole mass spectrometer (ThermoFisher Scientific, USA), was employed. Method limits of detection (MLODs) ranged from 0.013 to 0.038 ng mL^{-1} (Table S2); the detection frequencies are shown in Table S3.

Calf serum (Invitrogen, Oslo, Norway) not containing PFASs (background lower than the limit of detection) was used for method validation, quality control, and quality

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1 PFBA, 2 PFPeA, 3 PFHxA, 4 PFBS, 5 PFHpA, 6 PFOA 7 PFHxS, 8 PFNA, 9 PFHpS, 10 PFDA, 11 PFOS 12 PFUnDA, 13 6:2Cl-PFESA, 14 PFDoA, 15 PFDS, 16 PFTrDA, 17 8:2 Cl-PFESA, 18 PFOSA, 19 PFTeDA, 20 PFHxDA, 21 PFODA

Figure 1. Typical chromatogram of the standard mixture of 21 targeted PFASs (1 ng mL^{-1}, each).

assurance of the analytical method. Matrix spiked recoveries were measured by spiking at three levels (i.e., 0.05, 0.5, and 5 ng mL^{-1} of each PFAS), and the spiked recoveries of PFASs were ranging from 83.8% to 115%. To avoid background contamination, all of the flow tubes of the mobile phase of UltiMate 3000 DGLC were changed into polyproplene tubes and polyether ether ketone materials without PFASs background contamination. All containers and equipment were precleaned with LC-MS-grade methanol, and all solutions and reagents used in sample preparation and analysis were tested and confirmed that they did not contain PFAS. A procedural blank (i.e., MeOH) was used to monitor background PFASs and the possible carry-over between samples. A quality control sample (i.e., 2 ng mL^{-1}) was injected every eight samples to monitor the stability of peak intensity and the drift of retention time for PFASs. Chromatograms of the 21 targeted PFASs are shown in Figure 1. Detailed information about the analytical method used in the present study is presented in the Supporting Information (SI), and native standards and detection parameters are listed in Table S4.

Dissociation Constants (K_d) of the Protein (Serum)−PFAS Complexes. HSA and L-FABP are the most important transport proteins in human serum, and they were selected to study the relationship between K_d and PTEs. Human serum (containing all endogenous proteins) was regarded as a whole entity to calculate the $K_{d\text{-}SP}$ of PFASs in the present study. To verify the stability of $K_{d\text{-}SP}$ values in different sera, four human serum samples (with nondetectable or low PFASs) were assigned to calculate the individual K_d values of serum−PFAS complexes ($K_{d\text{-}SP}$). The $K_{d\text{-}SP}$ values were highly consistent in four different human serum samples (RSD ranging from 0.5% to 11.6%), suggesting that serum could be considered as a whole entity for obtaining properties, such as K_d (Figure S1). The $K_{d\text{-}SP}$ and K_d values of HSA−PFAS ($K_{d\text{-}HP}$) and L-FABP−PFAS complexes ($K_{d\text{-}LP}$) were obtained by an equilibrium dialysis, which is considered to be gold standard approach in protein binding studies.[40,44] The equilibrium dialysis experiment is based on a rapid equilibrium dialysis device with a molecular weight membrane cutoff of 8 kDa (Thermo fisher scientific, USA). This device needs 4 h to achieve equilibrium, and then precipitated buffer (namely, 90/10 acetonitrile/water with 0.1% formic acid) was used to precipitate the protein and release PFASs. The standard solution used for calculation of K_d for PFOS binding to proteins is pure linear solution, which is purchased from Wellington Lab (Canada). Detailed information about the equilibrium dialysis experiment is shown in the Supporting Information.

The calculation of K_d of the protein (serum)−PFAS complexes was derived from a previous method.[45,46] In the process of PFAS binding to the protein (serum), the free state PFAS (L_F) could bind reversibly to the protein (R) and form the protein (serum)−PFAS complex (LR), as shown in eq 1

$$L_F + R \Leftrightarrow LR \quad (1)$$

At equilibrium, K_d is an indicator for evaluating the affinity of PFAS binding to protein (serum) and is defined by eq 2

$$K_d = \frac{[L_F][R]}{[LR]} \quad (2)$$

Han et al. (2003) proposed an equation (eq 3) for linking L_F to K_d

$$\frac{[L]_F}{r} = \frac{[L]_F}{n} + \frac{K_d}{n} \quad (3)$$

where $[L]_F$ refers to the molar free state concentration of PFAS, r refers to the moles of PFAS binding to per mole of protein, and n refers to the number of binding sites; the units of [L], [R], and K_d are all in molar units (M). There is a linear association between $[L]_F/r$ and $[L]_F$, and $1/n$ and K_d/n are the slope and the intercept, respectively; then the K_d value can be calculated.

Statistical Analysis. Data processing was performed using R (version 3.4.1). For statistical analysis, we replaced values below the MLODs with MLODs divided by the square root of 2. To better estimate PTEs of PFAS, only paired samples with concentrations > MLOD in both maternal and cord serum were used. One-way analysis of variance (ANOVA) was used to compare the differences in the PTEs of PFASs, and linear regression was performed between PTEs and following individual predictor variables for each PFAS. Covariates were included one at a time: maternal age (continuous, year) and gestational age (continuous, weeks), categorical variable such as maternal weight (<66.8, 66.8−74.7, ≥74.7, kg), height (<1.60, 1.60−1.65, ≥1.65, m), prenatal BMI (<25.4, 25.4−

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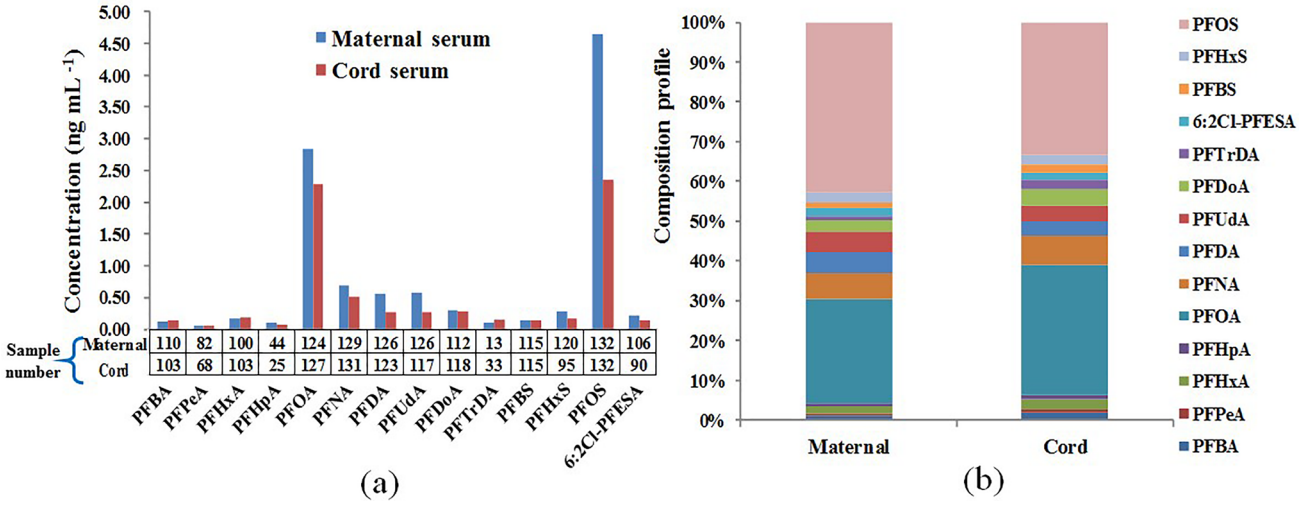


Figure 2. (a) Median concentrations of PFASs in maternal and cord serum, and (b) composition of PFASs in maternal and cord serum.

28.1, ≥28, kg/m^2), infant's sex (male, female), parity (no child, ≥one child), birth weight (<3.29, 3.29−3.68, >3.68), and birth length (<50, 50, >50). The categories of maternal weight, height, prenatal BMI, birth weight, and birth length were based on tertiles. Moreover, multivariate linear regression models were also established to explore associations between PFASs concentration and birth outcomes (including birth weight, birth weight, birth length, ponderal index, and gestational age), and PFAS levels were divided into low-, middle-, and high-exposure groups. For PFHpA and PFTrA, due to their low detection frequencies in serum, the participants were divided into a low-exposure group (<MLOD) and the remaining parts are divided into two groups (middle- and high-exposure groups) according to the concentration. From a statistical point of view, the multivariable model adjusted for confounding factors. Potential confounders, including maternal age, weight, and height, prenatal BMI, infant's sex, parity, and gestational age, were entered into the final multivariable model with $p < 0.20$ in univariate associations with birth outcomes. Statistical significance was defined as a $p < 0.05$. Exposure groups were included as ordered variables in the final multivariate regression model for trend testing. Moreover, statistically suggestive was defined as a $p < 0.10$, which was suggested by a previous study.[47]

RESULTS AND DISUSSION

Concentrations of PFASs in Maternal and Cord Serum. The detection frequencies and concentrations of PFASs in maternal and cord serum and number of paired samples with concentrations > MLOD are listed in Table S3, and the median value is shown in Figure 2. C8 PFASs, including PFOS and PFOA, were the dominant PFASs in both cord and maternal serum (Figure 2a). Short-chain PFAAs, including PFBA, PFPeA, PFHxA, PFHpA, and PFBS, were detected in 83%, 62%, 76%, 33%, and 87% of the maternal serum and 78%, 52%, 78%, 19%, and 87% of the cord serum, respectively, with median levels of 0.10, 0.05, 0.17, 0.04, and 0.14 ng/mL in maternal serum, and 0.12, 0.04, 0.18, 0.04, and 0.12 ng/mL in cord serum, respectively. PFOS and PFOA was defined as C8 compounds, and PFCAs with a carbon chain length > 8 (C9−C13 PFASs in the present study) was defined as the long-chain PFASs. Long-chain PFASs were detected in almost one-half of the serum samples, and PFNA was the major long-chain PFAS, which was detected in 98% and 99% of the maternal (median concentration 0.57 ng/mL) and cord serum samples (median concentration 0.38 ng/mL), respectively. Short-chain PFAAs (PFBA, PFPeA, PFHxA, PFHpA, and PFBS) and long-chain PFASs accounted for 5.68% and 20.1% of total PFASs in the maternal serum samples, while they accounted 9.12% and 20.9% of the total PFASs in the cord serum samples (Figure 2b).

A comparison of PFAS concentrations between the present study and previous literature is summarized in Table S5. Although PFOS was regulated around the world a decade ago, it was still the major PFAS detected in human serum.[48−50] The median levels of PFOS and PFOA (4.07 and 2.21 ng/mL in maternal serum and 1.80 and 1.88 ng/mL in cord serum, respectively) in the present study were lower than those reported in samples collected before 2009,[26,51] and these median values were comparable to those in samples collected during a similar period.[48−50] As a novel PFAS was only produced in China, 6:2 Cl-PFESA was reported in Chinese pregnant women before.[17,52] The median 6:2 Cl-PFESA value in the present study was 0.094 and 0.091 ng/mL in maternal and cord serum, respectively, which was slightly lower than those in samples collected from Wuhan in 2014 (1.89 and 0.80 ng/mL in maternal and cord serum, respectively;[17] 1.54 and 0.60 ng/mL in maternal and cord serum, respectively[52]). This might be due to the fact that Wuhan has been an important fluorine chemical base in China since the 1990s, and factories there might have been producing 6:2 Cl-PFESA for a long time.[53,54]

Associations between PFASs and Birth Outcomes. Associations among individual PFAS and birth outcomes of fetuses are shown in Table S6 and Table S7. PFBA in maternal serum (n = 110) showed significant negative association with birth length, with an estimated decrease of 0.60 cm (95% CI, −1.03 and −0.17 cm) in the highest exposure group than in the lowest exposure group (p for trend = 0.004), and PFBA in cord serum (n = 103) showed suggestive association with birth weight, PI, with p for trend = 0.07. PFHxS in cord serum (n = 95) showed suggestive association with gestational age, with an estimated decrease of 0.47 weeks (95% CI, −0.92 and −0.02 weeks) for the highest versus lowest exposure group (p for trend = 0.053). For long-chain PFASs, PFDoA in cord serum (n = 118) showed significantly association with birth length,

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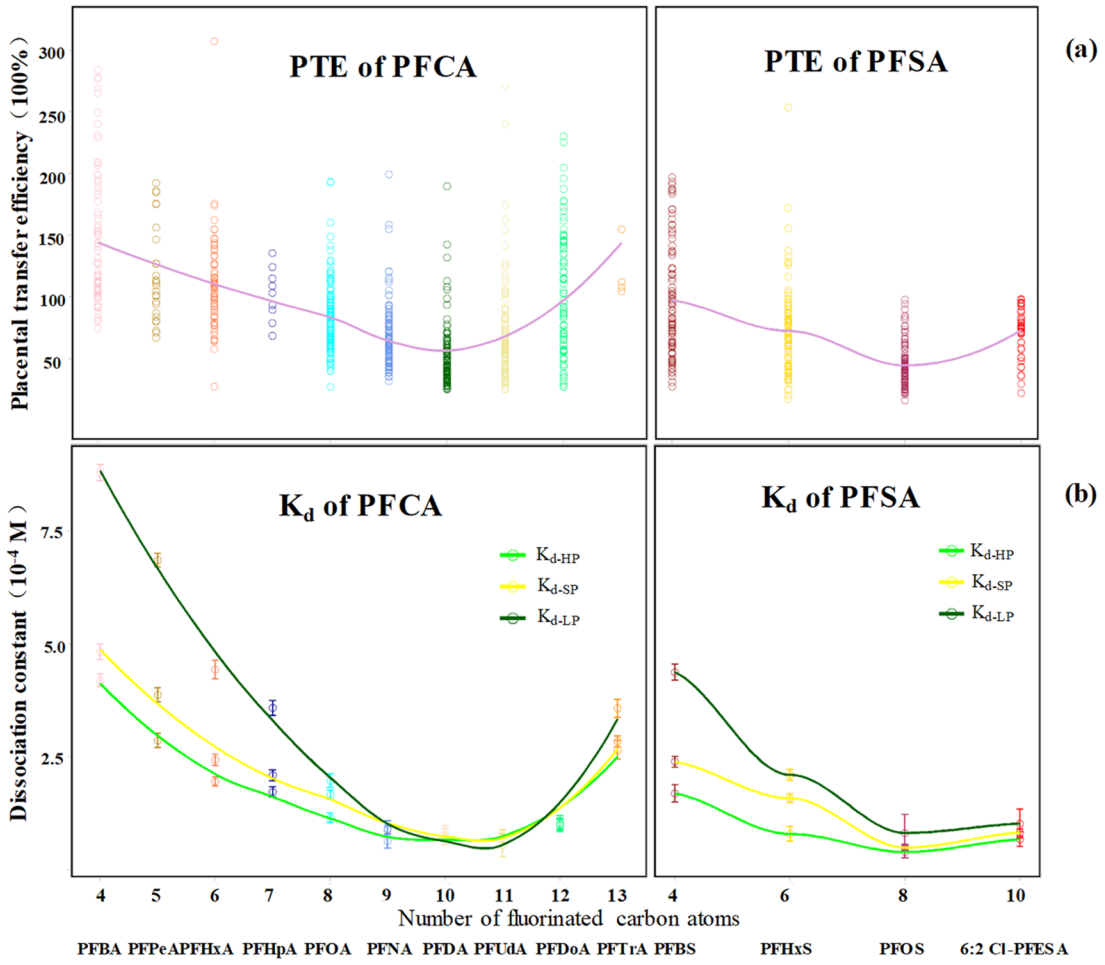


Figure 3. Trends of (a) PTEs of the PFASs (%) and (b) K_d of the PFASs (M) along with the chain length.

with an estimated increase of 0.67 (95% CI, 0.27 and 1.10 cm) and 0.50 cm (95% CI, 0.07 and 0.92 cm) for the middle versus lowest exposure group and highest versus lowest exposure group, respectively (p for trend = 0.018). PFTrA in cord serum (n = 33) showed significant positive association with birth weight, with an estimated increase of 263 g (95% CI, 52.9 and 474 g) for the middle-exposure group versus lowest exposure group (p for trend = 0.044). For the novel PFASs, 6:2 Cl-PFESA (n = 90) in cord serum showed significantly positive association with birth length, with an estimated increase of 0.54 cm (95% CI, 0.12 and 0.95 cm) for the highest versus lowest exposure group (p for trend = 0.016).

There were inconsistent associations between PFASs and birth outcomes reported in the literature,[24,55−60] and most studies focused on the related birth outcomes, such as birth weight, of PFOS and PFOA.[24,55,56,58,60] Birth weight did not associate with PFOS or PFOA in maternal r cord serum in the present study, which was consistent with the null associations reported in previous studies.[48,55] PFASs other than PFOS and PFOA were currently not regulated in the recent past, and birth outcomes affected by these compounds were studied to a lesser extent. It was noteworthy that PFBA (C4) in maternal serum was significantly negatively associated with birth length, and cord PFBA had suggestively positive association with birth weight (p = 0.07) and PI (p = 0.07) in our study. Testing on both human and animal tissues showed that short-chain PFAAs were distributed in various organs/tissues.[61−63] A few studies were carried out to study the toxicity of PFBA on animal models. Das et al.[64] reported that PFBA did not produce developmental toxicity during pregnancy in mice, while Mahapatra et al.[65] observed that localized lipid droplets were accumulated in PFBA-exposed cells and suggest that PFBA could affect cell metabolism. Chang et al.[66] suggested that PFBA could increase thyroid hormone turnover and transiently increase free T4. Moreover, PFDoA and PFTrA were observed to have a suggestive or significantly positive association with birth weight or length of fetuses in our present study. An animal experiment showed that exposing zebrafish larvae to PFDoA could cause disruption of the hypothalamus−pituitary−thyroid axis.[67] Kim et al.[15] reported PFTrA in both maternal and cord serum have significantly negative associations with fetal thyroid hormone level. The latter is suggested to be a critical factor for fetal growth and development; thus, the influence on birth outcomes related to these PFASs might be mediated through this pathway.

To our knowledge, this is the first study that reports PFBA and PFBS in paired maternal and cord serum, and both PFBA and PFBS might affect birth outcomes. The production of short-chain PFAAs has increased in the past several years due to the regulation of legacy long-chain PFAAs, and their levels in humans have therefore been increasing sera.[29,68] The possible adverse effects of short-chain PFASs need further attention, especially after long-term exposure, and more

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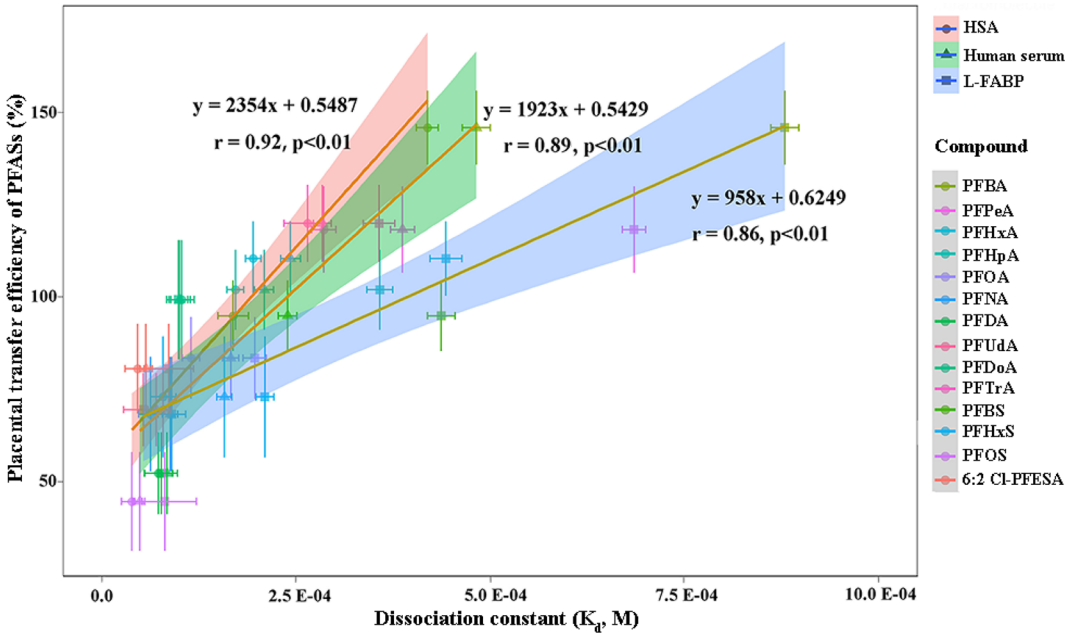


Figure 4. Correlation between the PTE and $K_{d\text{-}SP}$, PTE and $K_{d\text{-}HP}$, and PTE and $K_{d\text{-}LP}$.

attention should be paid on susceptible populations, such as fetus and children.

Placental Transfer Efficiencies (PTEs) of PFASs. The occurrence of PFASs in cord serum suggested that these chemicals can cross the placenta barrier, and the mean of PFASs PTEs ranged from 44% (PFOS) to 146% (PFBA). A U-shaped trend for the PTEs along with an increasing chain length was observed (the ether linkage contained in the backbone of 6:2 Cl-PFESA was counted in the chain length, and 6:2 Cl-PFESA considered one molecule longer than PFOS) (Figure 3a), which was in line with previous studies.[16,17,28] To the best of our knowledge, there has been no information on PTEs of short-chain PFCAs (C4−C6) to date; the PTE of PFHpA (C7) was the shortest chain length PFAS ever reported in the literature.[16,17] They were neglected because of their lower toxicity in animal experiments or their concentrations being under MLODs, which might be due to the low sensitivity of the analytical methods. PTEs had a descending trend from PFBA (C4, PTE = 146%) to PFDA (C10, PTE = 44%) and then started to increase with increasing carbon chain length from PFDA to PFTrDA (C13, PTE = 120%). The PTEs for short-chain PFAAs, such as PFBA (146%), PFPeA (118%), and PFHxA (110%), were much higher than those of legacy PFASs, such as PFOA (83.2%) and PFOS (44%). Moreover, we found that the mean values of PTEs for C4−C7 PFCAs and C13 PFCA) were all above 100%, indicating these compounds are more inclined to accumulate in the fetus. PFCAs tend to transfer more efficiently than PFSAs with the same fluorinated chain length; for example, PFPeA had a mean PTE of 118%, 1.2-fold higher than that of PFBS (97%); PFHpA had a mean PTE of 102%, 1.4-fold higher than that of PFHxS (73%). Our present study provided PTEs of C4−C6 PFCAs, which filled the data gaps in PTEs for short-chain PFAAs.

To date there has been little information available on short-chain PFAAs in pregnant women and fetuses, not to mention the PTEs. Among human monitoring studies, Olsen et al.[69] found that the mean serum elimination half-life of PFBS was only 25.8 days in occupational workers, much shorter than the shortest half-life of PFOS (1.9 years), PFOA (1.7 years), and PFHxS (3.6 years) ever reported.[54] Short-chain PFAAs were also considered nonbioaccumulative in the environment.[5,70] Due to these properties of short-chain alternatives, the production dramatically increased in recent years. As a result, they were ubiquitously detected in environmental studies and the general population.[30,70,71] Our present study implied a widely distribution of short-chain PFAAs in humans, and they could be transferred to the fetus more effectively than legacy PFAAs. Although animal experiments and field investigations have suggested low bioaccumulative potentials for short-chain PFAAs in aquatic ecosystems,[70] their higher PTEs increased the risk to susceptible populations, such as fetuses.

Associations between Demographic Characteristics and PTEs. Demographic characteristics such as mother age and weight were considered confounding factors for chemical-specific PTEs of PFASs in previous studies.[17,72] Associations between demographic characteristics and PTEs in the present study were carefully evaluated (Table S8). A statistically significant association between the maternal weight and PTE of PFOS was observed, with an estimated increase of 10.1% (95% CI, 2.3% and 17.9%) for the second tertile versus lowest tertile group (p for trend = 0.03). Besides, PTE of PFNA showed suggestive association with maternal weight (p for trend = 0.07). Women with high BMI tended to increase PTE of PFDA, with an estimated increase of 0.4% (95% CI, −7.9% and 8.7%) and 12.8% (95% CI, −0.9% and 26.6%), respectively, for the second tertile and third tertile versus lowest tertile group (p for tend = 0.04). Moreover, older mothers tended to decrease PTE of PFUdA, with a decrease of 2.6% (95% CI, −4.6% and −0.5%) (p = 0.015) per year increase in maternal age, while PTE of PFUdA showed suggestively positive association with birth length, with an estimated increase of 29.1% (95% CI, −0.4% and 57.8%) for the third tertile versus the first tertile group (p for tend = 0.06). The present study showed that maternal age, maternal weight, and BMI could influence the PTEs of PFASs, which was consistent with previous studies.[17,72] The reason might be possibly that maternal age and weight would affect the

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development of placental vascular, consequently interfering the function of placenta.[73]

Association between K_d and PTEs. The literature has shown that PFASs could bind to proteins, such as serum albumin and L-FABP, and the affinities showed an inverted U-shape with increasing chain length.[42,74] To our knowledge, no experimental evidence has demonstrated the association between the binding affinities of PFASs to serum and the PTE.

The $K_{d\text{-}SP}$, $K_{d\text{-}HP}$, and $K_{d\text{-}LP}$ of individual PFASs are listed in Table S9, and $K_{d\text{-}SP}$, $K_{d\text{-}HP}$, and $K_{d\text{-}LP}$ also showed a U-shaped trend along with increasing perfluoroalkyl chain length (Figure 3b). PFBA (C4) had the highest $K_{d\text{-}SP}$, $K_{d\text{-}HP}$, and $K_{d\text{-}LP}$, and they were 482 ± 45, 419 ± 37, and 879 ± 51 μM, respectively. $K_{d\text{-}SP}$ and $K_{d\text{-}HP}$ decreased from C4 to C10 and then increased from C10 to C13. However, $K_{d\text{-}LP}$ decreased from C4 to C11 and then increased from C11 to C13. The $K_{d\text{-}HP}$ and $K_{d\text{-}LP}$ values reported in our study were similar to those reported in previous studies (Table S9). Information on the PFAS-HSA interaction mainly focused on PFOS and PFOA, and the $K_{d\text{-}HP}$ values of PFOA and PFOS generally ranged from 10 μM to 1000 μM and from 1 μM to 100 μM in previous studies,[42,44−46,75] respectively. Moreover, the lowest $K_{d\text{-}HP}$ of linear PFOS ever reported was only 8×10^{-2} μM.[46] The large range of dissociation constants for PFASs might be due to methodological differences, such as equilibrium dialysis,[44] fluorescence,[75] ultrafiltration centrifugation,[46] size exclusion chromatography,[45] and the purity of PFASs standards (pure linear vs technical PFASs) and protein (dry powder vs solution), but the trends of the dissociation constants for individual PFASs were generally consistent in previous studies. We reported the complete $K_{d\text{-}HP}$ and $K_{d\text{-}LP}$ values of PFCAs with a carbon length ranging from C4 to C13. Except for PFUdA and PFDoA, the affinity of PFAS binding to HSA was greater than that to L-FABP. Consistently, Zhang et al.[42] also found PFUdA to have the highest affinity (10.6 ± 1.2 μM) to L-FABP among PFCAs with 7−13 fluorinated carbons. The $K_{d\text{-}SP}$ values for individual PFASs were between the corresponding $K_{d\text{-}HP}$ and $K_{d\text{-}LP}$ values. Spearman correlation analysis was carried out to explore the relationship between the PTE and the K_d of the same PFAS. $K_{d\text{-}SP}$, $K_{d\text{-}HP}$, and $K_{d\text{-}LP}$ were all significantly associated with the PTE ($p < 0.01$) (Figure 4).

PFASs might pass through the placenta barrier by passive diffusion;[17,43] hence, this process was influenced by the free PFASs concentration gradient between both sides of the placental barrier. The higher K_d values mean weaker binding affinities to proteins in serum, which might cause a higher free fraction of these PFASs in serum. Short-chain and long-chain PFAAs with relatively high K_d could transfer from maternal to cord more efficiently. The significant correlations between K_d and the PTE generally supported our hypotheses that K_d dominated the placental transfer process of PFASs. The correlation coefficients of the fitted curves between $K_{d\text{-}SP}$ and the PTE, $K_{d\text{-}HP}$ and the PTE, and $K_{d\text{-}LP}$ and the PTE were 0.89, 0.92, and 0.86, respectively, which indicated $K_{d\text{-}HP}$ was most relevant to PTEs of PFASs. Moreover, the slopes of the $K_{d\text{-}HP}$-PTE curve (2354) were also biggest compared to the $K_{d\text{-}SP}$-PTE curve (1923) and $K_{d\text{-}LP}$-PTE curve (958), which further suggested the $K_{d\text{-}HP}$ was the critical factor and regulated the placental transfer process of PFASs, and this conclusion might be extended to the placental transfer process of other pollutants. Additionally, the blood composition such as packed cell volume (PCV) and albumin concentration in maternal blood and cord blood were not consistent.[76] The newborns have a higher PCV (0.62) than pregnant women (0.38);[77] however, PFASs were mainly distributed in serum. Therefore, the different PCVs between the mothers and the fetuses might not be the important factor influencing the PFASs transfer from maternal blood to cord blood. The albumin concentration in cord serum is lower than that in maternal serum,[78] which might cause the competition for PFASs between the albumins on each side of the placenta.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.est.9b00715.

Online Turboflow HPLC-MS/MS procedure for PFASs analysis; method validation, quality control, and quality assurance (QC/QA); steps for incubation and equilibrium dialysis and calculation of $K_{d\text{-}SP}$, $K_{d\text{-}HP}$, and $K_{d\text{-}LP}$; demographic characteristics of studied population; method limits of detection and quantitation of targeted PFASs in the present study; supplier, purity of targeted PFASs standards and internal standards, solvents, and protein used in the present study; concentrations of PFASs in maternal and cord serum; summary of PFASs median concentrations in the present study and previous studies; multiple linear regression coefficients between PFAS concentrations in cord serum and birth outcomes; multiple linear regression coefficients between PFASs concentrations in cord serum and birth outcomes; percentage change in PFASs PTEs associated with individual factors; $K_{d\text{-}SP}$, $K_{d\text{-}HP}$, and $K_{d\text{-}LP}$ in the present and previous studies; $K_{d\text{-}SP}$ in four different serum samples (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*Tel.: +86-10-6284-9357; E-mail: jjfu@rcees.ac.cn.

ORCID

Jianjie Fu: 0000-0002-6373-0719
Aiqian Zhang: 0000-0001-5680-2529
Maoyong Song: 0000-0002-6776-4705
Guibin Jiang: 0000-0002-6335-3917

Author Contributions

⊥K.G. and T.Z.: These authors contributed equally to this work

Notes

The authors declare no competing financial interest.

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